Computers in Biology and Medicine 187 (2025) 109768



Contents lists available at ScienceDirect

Computers in Biology and Medicine

journal homepage: www.elsevier.com/locate/compbiomed





A comparative study of statistical, radiomics, and deep learning feature extraction techniques for medical image classification in optical and radiological modalities

Pegah Dehbozorgi [a,b], Oleg Ryabchykov [a,b], Thomas W. Bocklitz [a,b,*]

[a] Leibniz Institute of Photonic Technology, Member of Leibniz Health Technologies, Member of the Leibniz Centre for Photonics in Infection Research (LPI), Albert-Einstein-Strasse 9, 07745, Jena, Germany
[b] Institute of Physical Chemistry (IPC) and Abbe Center of Photonics (ACP), Friedrich Schiller University Jena, Member of the Leibniz Centre for Photonics in Infection Research (LPI), Helmholtzweg 4, 07743, Jena, Germany

ARTICLE INFO

Keywords:
Medical image classification
Feature extraction
Machine learning
Radiomics

ABSTRACT

Feature extraction in ML plays a crucial role in transforming raw data into a more meaningful and interpretable representation. In this study, we thoroughly examined a range of feature extraction techniques and assessed their impact on the binary classification models for medical images, utilizing a diverse and rich set of medical imaging modalities. Using H&E-stained, chest X-ray, and retina OCT images, we applied methods to extract statistical, radiomics, and deep features. These features were then used to develop PCA-LDA models as the employed classifier. We evaluated the models based on two decisive metrics: latency and performance. Latency measured the time taken for feature extraction and prediction, while mean sensitivity (balanced accuracy) characterizes the model performance. Our comparative study revealed that statistical and radiomics features were less effective for medical image classification, as they showed high latency and lower performance scores. In contrast, pre-trained DL networks performed efficiently, with high sensitivity and low latency. For H&E-stained images, the statistical feature extraction took about an hour and achieved 90.8 % sensitivity, while ResNet50 reduced processing time fourfold and increased sensitivity to 96.9 %. For chest X-rays, radiomics features were time-intensive with 92.2 % sensitivity, while ResNet50 improved sensitivity to 96 % with faster extraction time. For retina OCT images, radiomics yielded a sensitivity of 91 %, while DenseNet121 achieved 98.6 % sensitivity in 15 min. These findings underscore the superior performance of DL techniques over the statistical and radiomics features, highlighting their potential for real-world applications where accurate and rapid diagnostic decisions are essential.

1. Introduction

Diagnosis involves identifying a disease or illness through a comprehensive patient examination. Unfortunately, many diseases and conditions affect areas of the body that remain unseen to the naked eye. Diagnostic medical imaging serves as a reliable tool in this process, enabling the visualization of internal abnormalities and facilitating accurate diagnosis. As a diagnostic tool, medical imaging employs technology to observe the human body, assisting in diagnosing, monitoring, and treating medical conditions. Visualizing structures concealed by skin and bones holds significant importance in the control of diseases. Traditionally, the human visual system is responsible for image processing and perception. When analyzing medical images to diagnose medical conditions, doctors have long relied on the visual system to do the job. However, the trend towards increasingly detailed medical images threatens to overwhelm the human visual system [1]. Manual image analysis methods are labor intensive and susceptible to both intra- and inter-observer variability. Automated medical image analysis techniques offer a solution to these limitations [2].

Recently, there has been a notable surge in employing Machine Learning (ML) methods for medical image analysis, primarily focusing on improving efficiency and reducing potential human errors. However, the rapid advancement of Deep Learning (DL) as a subfield of ML has led to a significant shift towards harnessing this powerful tool to enhance the accuracy of medical image analysis even further [3]. These ML and DL methods involve learning and extracting meaningful features from

* Corresponding author. Leibniz Institute of Photonic Technology, Member of Leibniz Health Technologies, Member of the Leibniz Centre for Photonics in Infection Research (LPI), Albert-Einstein-Strasse 9, 07745, Jena, Germany.
E-mail address: Thomas.bocklitz@uni-jena.de (T.W. Bocklitz).

https://doi.org/10.1016/j.compbiomed.2025.109768
Received 28 August 2024; Received in revised form 9 December 2024; Accepted 28 January 2025
Available online 31 January 2025
0010-4825/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

medical images, which can then be used to make predictions and establish classification tasks for disease detection.

In that context image features are essential for addressing specific applications and hold significant properties of images [4]. In the context of medical imaging, a "feature" refers to a specific characteristic or pattern within an image that is relevant to the diagnostic task. These features can encompass various aspects, including shape, size, texture, density, or intensity of structures within the image. Image processing might involve extracting features to create a more informative dataset which can be further utilized for various tasks such as identification, classification, diagnosis, and detection.

In this study, we undertake a systematic examination of various classical ML and advanced DL feature extraction techniques to better understand how different approaches impact classification outcomes. In our investigation, we primarily focused on two evaluation metrics: latency and performance. Latency evaluation measures the time taken by the model to process and complete a specific task, namely feature extraction and prediction. This metric is crucial in dynamic applications where quick responses are essential. Understanding and optimizing latency helps ensure that the model can respond promptly to new data inputs without significant delays, thereby enhancing overall system efficiency and user experience. Performance evaluation, on the other hand, is based on the mean sensitivity/balanced accuracy of the classification model. Sensitivity, also known as recall or true positive rate, quantifies the ability of the model to correctly identify positive instances out of all actual positive instances in the dataset. It provides insight into how well the model captures relevant patterns and discriminates between different classes or categories. High sensitivity score indicates that the model is effective at minimizing false negatives, which is critical in applications where missing positive instances can lead to significant consequences, especially in medical diagnostics.

This investigation provides an extensive and systematic comparison of feature extraction techniques, including statistical, radiomics, and DL-based methods, applied across three widely used medical imaging modalities: H&E-stained histological images, chest X-rays, and retina OCT scans. The main contribution of our work lies in utilizing a rich subset of medical image modalities to ensure generalizability. Building upon our analysis workflow, data collection, model development, and evaluation strategy, we conducted a focused analysis of different feature types and their impact on the classification outcomes. We thoroughly assessed the accuracy of the extraction techniques and placed a strong emphasis on evaluating their performance, particularly the time estimations for feature extraction and prediction. This study provides robust and reliable insights into how each method will likely perform in practical, real-world medical image analysis. The dual focus evaluation plan fills an important gap in the current literature, where applications like medical diagnostics require not only high accuracy but also prompt processing to support real-time decision-making. To the best of our knowledge, this study is the first to explore the impact of different feature extraction methods on the binary classification outcomes across diverse medical imaging modalities, focusing on both performance metrics, sensitivity values, and latency. This unique approach provides valuable insights for future analyses in the field.

The analysis has been established by collecting data and investigating three different datasets, which have included the most commonly used image modalities in healthcare specifically pathology, ophthalmology, and radiology domains for diagnosis and treatment purposes. In our earlier work [5], we noted that while pre-processing can be beneficial and improve the analysis results in some cases, using the wrong steps or strategies can potentially hinder the classification performance. We also discovered that, although pre-processing can boost the classification accuracy in specific situations, the potential for significant improvements is often limited. These observations encouraged us to conduct the current analysis using the raw format of the medical images.

1. The first dataset has been selected from a publicly available repository comprising H&E-stained patches extracted from the histological images of human colorectal cancer (CRC) and normal tissues [6]. The hematoxylin and eosin staining technique (H&E-stain) is a key tool in histology. It is widely used for its important role in medical diagnosis, particularly as the gold standard for detecting cancer. Pathologists routinely examine biopsy tissues for suspected cancer using H&E-stained histological sections due to their excellent clarity in visualizing cellular and tissue structures.
2. The second set of data consisting of chest X-ray images of humans classified into the two groups: those with normal conditions and those with pneumonia [7]. X-ray, a rapid and painless imaging technique, is frequently employed to visualize internal body structures. Primarily utilized for examining bones and joints, X-ray can also expose issues affecting soft tissues. Conditions like breast cancer and lung diseases such as pneumonia and lung cancer are among those detectable through X-ray imaging.
3. Finally, this study has integrated retinal Optical Coherence Tomography (OCT) scans as the third dataset. These scans of the adult patients, are comprising two distinct categories: choroidal neovascularization (CNV) and normal retina [7]. OCT is a crucial tool for healthcare providers in diagnosing and treating various eye conditions. It plays an important role in identifying disorders such as glaucoma and macular degeneration, aiding in their accurate diagnosis and management.

The study phases have been formulated as follows: feature extraction has been carried out utilizing various classical ML for statistical feature extraction and advanced DL techniques to extract deep features, followed by the application of Principal Component Analysis (PCA) [8] to reduce features dimensionality. The final phase has involved the development of predictive models through the implementation of Linear Discriminant Analysis (LDA) [9] and lastly the classification. This choice helped reduce the need for extensive hyperparameter tuning and optimization, allowing us to remain focused on assessing the contributions of statistical, radiomics, and deep features on the classification outcomes. Analyzing a rich subset of medical images and employing common feature extraction methods enables our study to guarantee its generalizability and facilitates drawing robust conclusions.

The "Related works" section provides a brief review of studies that explore and evaluate various feature extraction techniques, along with analyses conducted using different imaging modalities. The "Materials" section offers a comprehensive overview of the datasets, followed by the "Feature extraction and classification" section, which details the techniques that have been used for feature extraction and model development. The findings are presented and discussed in the "Results" and "Discussion" sections. The key findings of the study are summarized in the "Conclusion" section for clarity and to highlight key takeaways. The "Limitations" section also provides a brief discussion of the study's constraints."

2. Related works

A variety of classical ML and advanced DL techniques are employed for feature extraction, each designed to suit particular types of data and tasks. The Dharmender et al. [10], explored the challenges of ultrasound imaging in healthcare settings. Their study proposed the application of pre-processing techniques to improve image quality while preserving crucial information, as well as feature extraction focusing on second-order statistical texture features derived from the classical ML method, gray level co-occurrence matrix (GLCM) [11]. Additionally, dimensionality reduction was conducted using PCA to prepare the classification model for kidney ultrasound images. The research found that integrating GLCM with PCA resulted in notable classification accuracy when employing Artificial Neural Network (ANN) [12] for image classification. The Dheepak et al. [13], proposed a novel methods for

quantitative analysis of brain tumor images by integrating ML methods for feature extraction, GLCM and Local Binary Pattern (LBP) [14,15] features. When employing a linear support vector machine (SVM) [16] as the classifier, the combination of GLCM and LBP features offers a comprehensive representation of texture characteristics, leading to a notable enhancement in tumor detection and classification precision, achieving an impressive accuracy of 99.84 %. L Kanya Kumari et al. [17], addressed the significance of early stage detection of breast cancer using mammographic imaging. The study introduced an innovative technique known as the Advanced Gray-Level Co-occurrence Matrix (AGLCM) for extracting features from pre-processed images. The proposed framework encompassed image acquisition, pre-processing through Contrast Limited Advanced Histogram Equalization (CLAHE) [18], feature extraction using AGLCM, and classification employing the eXtreme Gradient Boosting (XGBoost) [19] classifier. Experimental results indicate promising outcomes with the combination of CLAHE, AGLCM and XGBoost methodology. The authors in Ref. [20] addressed the significant need for automating the classification of brain tumor types to enhance patient survival rates. In this study, the authors explored two distinct feature extraction methods, nLBP and αLBP, to categorize the most general brain tumor types. The nLBP method relied on pixel relationships, while the αLBP operator calculated pixel values based on angle values. The classification process contained the utilization of the ML algorithms, including K-Nearest Neighbor (KNN) [21], ANN, Random Forest (RF) [22], A1DE [23], and LDA. These algorithms were applied to the feature matrices extracted from the images using nLBP, αLBP, and classical LBP methods. The results presented promising performance, with the nLBP feature extraction method combined with the KNN model yielding the highest success rate in brain tumor classification. Additionally, radiomics, the extraction of quantitative features from medical images, has remarkable potential to drive advancements in precision medicine, especially in the field of oncology. In this study [24], Stefano explored the use of radiomics in PET imaging to improve tumor characterization and assess treatment responses across various cancer types. The study highlights that to fully unlock the potential of radiomics in improving patient care, future studies must focus on standardization, careful validation, and clinical integration in image analysis. Qing-qing et al., in Ref. [25] investigated the utilization of **computed tomography** (CT) radiomics features for pulmonary nodules classification. Radiomics feature models based on the CT images could well reflect the differences between benign nodules, malignant non-invasive nodules, and invasive nodules and assist in their classification across all the scenarios.

DL, mainly through the implementation of CNNs, delivers significant healthcare potential. CNNs help automate and improve diagnostic tasks by identifying and extracting important key features from medical images, making the analysis procedure faster and more accurate. In Ref. [26], the KONet model introduces a weighted ensemble learning approach that combines multiple CNN architectures together with feature extraction to improve classification accuracy. Whangbo et al. underscored the critical role of Artificial Intelligence (AI) in clinical practices, where accurate and fast diagnoses of knee osteoporosis (KOP) are essential for informed decision-making and reducing dependency on expertise. The analysis methods must not only deliver precise results but also operate efficiently for real-world medical applications. Proactively exploring such approaches is necessary to develop tools that meet the demanding accuracy and speed standards required in healthcare. Han et al. in Ref. [27] tried to address the challenge of predicting overall survival in patients with high-grade gliomas, a form of brain cancer with historically poor survival rates. The study introduced a novel approach using a combined DL and radiomics model to predict survival outcomes in a diverse group of patients. The model successfully classified patients into long- and short-term survivor groups by analyzing radiomics features and deep features extracted from brain MR imaging. The Dehbozorgi et al. utilized a pre-trained network, VGG16 [28], for feature extraction from medical images. The authors employed transfer learning as the central technique for feature extraction while investigating the effects of various pre-processing techniques on the classification models [5]. In Ref. [29], Ali et al., developed an automated method for detecting bacterial susceptibilities to three antibiotics using a combination of image processing and ML algorithms. They built their comparative analysis upon the first-order statistics-based features of the intensity and the bottleneck features of the trained U-Net network which were then used to train a one-class classification model. In Ref. [30], the study introduced Hybrid-NET, a DL framework that integrated DenseNet169 with ML classifiers, for classifying brain tumors. Using MRI scans, Hybrid-NET addressed the challenges of small dataset sizes and tumor complexity by leveraging feature extraction from DenseNet169 and combining predictions through a majority vote ensemble for enhanced accuracy. The model achieved a classification accuracy of 95.10 %, showing robustness in multiclass brain tumor detection through effective feature extraction with transfer learning. Khan et al. proposed an oral cancer detection model that utilized a feature-level fusion strategy to integrate features from EfficientNetB0 [31] and EfficientNetB1 [31], incorporating a novel self-attention mechanism. This approach improved detection performance by effectively combining features from both networks and benefiting from transfer learning during feature extraction. The self-attention block further enhanced interpretability, enabling the model to achieve a high accuracy of 98.83 % [32]. This research [33] demonstrated that combining deep features extracted from medical images with ML classifiers effectively improved diagnostic accuracy for disease differentiation. A novel DNLR-NET model was introduced for automated and precise Mpox diagnosis. The DNLR-NET model leveraged DenseNet201 for deep feature extraction, followed by logistic regression for classification, achieving a high accuracy of 97.55 %, thus proving the effectiveness of integrating deep features for disease detection.

Despite comprehensive research into feature extraction techniques, which includes classical ML and DL methods, identifying the optimal method for model development continues to pose a challenge. The diversity of data types and the varying complexity of problems mean that no single technique can be universally applied with consistent success. Researchers may choose to employ a specific method for feature extraction based on the nature of the problem, available data, and analysis objectives. The decision-making process is further complicated by factors such as the quality and quantity of data, the computational resources at hand, and the specific performance metrics of interest. Additionally, the rapid evolution of technology and algorithms continually reshapes the landscape, requiring ongoing adaptation and experimentation. Thus, the selection of feature extraction techniques is not only a matter of current best practices but also one of anticipating future developments and trends in the field.

3. Materials

Generalizability and robustness are two critical factors driving our approach. To enhance these aspects, we have incorporated a diverse range of image modalities, including the most commonly used ones in healthcare. By exploring various modalities, we aim to capture a broader range of diagnostic information. To ensure consistency in the analysis and fair comparisons for the classification outcomes, we deliberately resized all the images to a uniform dimension of 224x224 pixels. This standardized approach not only enhances the reliability of our results but also ensures that the image data meets the necessary input size requirement for the feature extraction phase when utilizing pre-trained networks.

The detailed specifications of each dataset will be thoroughly discussed in the subsequent sections.

3.1. Hematoxylin and Eosin stain (H&E-stained) dataset

Tissue diagnostic methods are employed to identify particular

structures and constituents in the preserved tissues. Staining enhances contrast by creating distinct colors and intensities in various components of the sample, thereby making structures within the tissue visible. The widely preferred stain combination is Hematoxylin and Eosin, abbreviated as H&E staining. H&E staining is the most commonly used technique for the disease diagnosis. It has been applied in nearly all clinical cases, and accounts for about 80 % of human tissue staining performed worldwide [34]. The initial dataset in our study comprises image patches extracted from human colorectal cancer (CRC) tissues and normal tissues, all stained with the hematoxylin and eosin (H&E) technique. These image patches were manually extracted from 86 H&E-stained human colorectal cancer tissue slides, obtained from formalin-fixed paraffin-embedded (FFPE) samples housed in the NCT Biobank (National Center for Tumor Diseases, Heidelberg, Germany) and the pathology archive of the University Medical Center Mannheim (UMM, Mannheim, Germany). Each patch has dimensions of 224 × 224 pixels and falls into one of two categories: normal (NORMAL) tissues or colorectal adenocarcinoma epithelium tumor (TUM) tissues.

3.2. Chest X-ray dataset

In principle, during the X-ray imaging [35] tissues absorb photons emitted by X-ray beams in relation to their electron density. This absorption varies according to the material's density. As the X-ray photons pass through the body, they are either captured by traditional film or modern image detectors. These detectors convert the body's attenuation of photons into digital images. These images provide a two-dimensional representation of a three-dimensional structure. Widely used in medicine, X-ray imaging helps diagnose conditions by visualizing bones, organs, and other internal structures [36]. In addition to the H&E images, the chest X-ray images are included as another under study dataset [7], broadening the scope of the analysis. These images were retrospectively selected from cohorts of pediatric patients aged one to five years at Guangzhou Women and Children's Medical Center, Guangzhou. All X-ray imaging was conducted as part of routine clinical care. For analysis, the images underwent an initial quality control process, where low-quality or unreadable scans were excluded. Two experts independently labeled the remaining images before the data were approved for training the AI system. To further minimize grading errors, a third expert reviewed the evaluation set, ensuring a high level of accuracy. This dataset involves normal chest X-ray images showing clear lungs (NORMAL), as well as images indicating pneumonia (PNE) in the lungs, all sized 224 × 224 pixels.

3.3. Optical Coherence Tomography (OCT) dataset

Optical Coherence Tomography (OCT) [37] is an imaging approach that produces high-resolution, real-time images in various dimensions, including one-dimensional depth, two-dimensional cross-sections, and three-dimensional volumes. OCT offers micron-level details, enabling a deeper look into the tissue under examination. This modality captures structural data by analyzing the light reflected from different layers within the sample. It is a widely employed diagnostic tool in healthcare, with a particular focus on imaging the layers of the retina in the eye [36]. The last image modality integrated into the study comprises optical OCT images obtained from adult patients. These images are categorized into two groups: normal OCT images depicting a healthy retina without any retinal fluid or edema (NORMAL) and images displaying indications of choroidal neovascularization (CNV). All images share the same pixel dimensions of 224 × 224 [7]. These scans were sourced from adult patients at the Shiley Eye Institute at UC San Diego, the California Retinal Research Foundation, Medical Center Ophthalmology Associates, Shanghai First People's Hospital, and Beijing Tongren Eye Center, covering the period from July 1, 2013, to March 1, 2017. For the training of AI models to be accurate, every image underwent a thorough grading process involving multiple levels of trained graders. The first level involved undergraduate and medical students who had completed an OCT interpretation course, and they conducted the initial quality control step by eliminating images with severe artifacts or low resolution. The remaining images were then independently graded by four ophthalmologists at the second level. Finally, the process was concluded by two senior retinal specialists, each with over 20 years of experience, who verified the final labels for each image.

Fig. 1, illustrates an overview of the datasets investigated in this study, all standardized to the dimension of 224 × 224 pixels. Subfigure (a) showcases examples of the H&E-stained images categorized into two classes: NORMAL for normal tissues and TUM for tumorous tissues. Subfigure (b) exhibits examples of the chest X-ray images representing NORMAL for normal lung images and PNE for lung images with pneumonia. Subfigure (c) displays examples of the OCT images representing NORMAL for normal retina images without any retinal fluid/edema and CNV for images displaying signs of choroidal neovascularization. Furthermore, subfigures (a), (b), and (c) offer a comprehensive overview of the datasets across both training and test sets, revealing the distribution strategy employed for model development and evaluation.

4. Feature extraction and classification

This study has been conducted through several steps, beginning with the collection of data from open repositories and selecting the most widely used modalities in the healthcare system. The data collection process has been followed by the feature extraction step. Feature extraction involves converting raw data into numerical features that retain the essential information from the original dataset, enabling further processing. It identifies the most distinctive characteristics within the data, facilitating easier consumption by ML or DL algorithms. Feature extraction aims to decrease data complexity, often referred to as data dimensionality while preserving as much relevant information as possible. This enhances the performance and efficiency of ML algorithms and simplifies the analysis process.

Given the significance of features in ML and DL tasks, this investigation primarily focuses on employing a variety of feature extraction methods. These methods range from classical ML techniques, which extract statistical features, to advanced DL models, which extract deep features. Our goal is to design the predictive binary classification models using the features derived from aforementioned techniques. We will compare the results and evaluate the efficacy of these methods based on various criteria, including the time required for feature extraction and prediction, as well as overall model performance through mean sensitivity evaluation. As mentioned earlier, performing a comprehensive and systematic study [5], we have demonstrated that while pre-processing can indeed enhance the classification performance in specific instances, the potential for substantial changes and improvements remains relatively limited. Regarding this observation and the current objective of the study, we have opted to skip the pre-processing step and utilize the images in their raw format. The feature extraction methods have been categorized into three distinct groups, as detailed below:

1. Statistical features include those extracted using methods such as mean, standard deviation, LBP, GLCM, and Histogram of Oriented Gradients (HOG) [38].
2. Radiomics features extracted using a commonly known open-source Python package for the extraction of radiomics features [39] from the medical images.
3. DL approach utilizing five pre-trained networks, VGG16, ResNet50 [40], DenseNet121 [41], MobileNetV2 [42], and InceptionV3 [43], as the feature extractors

With the extracted features, we proceeded towards the development of basic PCA-LDA model as our implemented classifier. Finally, we could thoroughly analyze the classification results and estimate the time

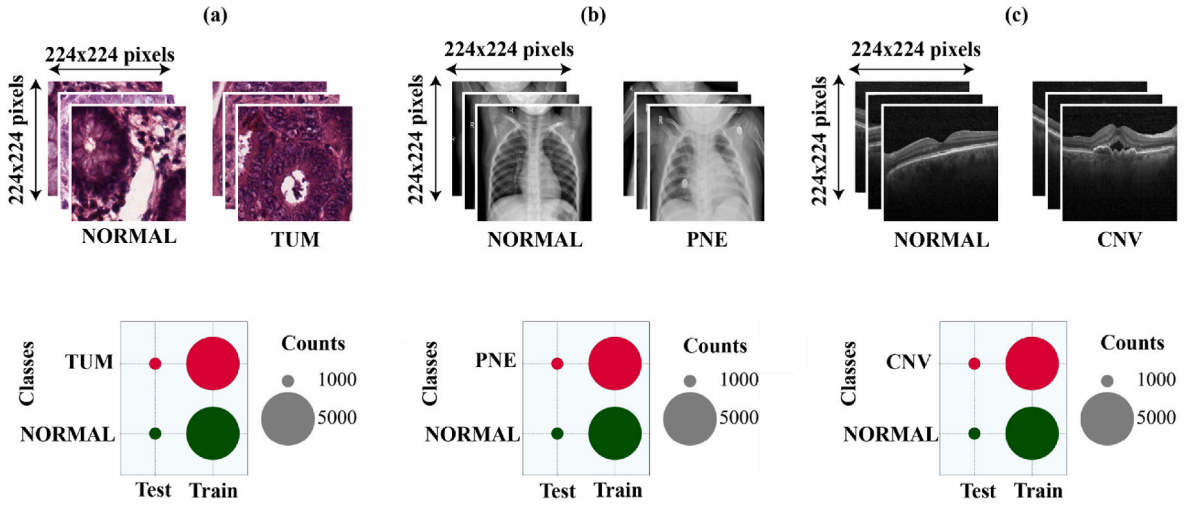


Fig. 1. Characteristics of the analyzed images across each dataset. This figure offers an overview of the images studied within each dataset. For consistency, all the images have been resized to 224 × 224 pixels. Subfigure (a) shows examples of the H&E-stained images from two categories: normal tissues labeled as NORMAL and tumorous tissues labeled as TUM. Subfigure (b) illustrates examples of the chest X-ray images from two categories: normal lung images labeled as NORMAL and lung images with pneumonia labeled as PNE. Subfigure (c) depicts examples of the retina OCT images from two categories: normal retina images without retinal fluid/edema labeled as NORMAL and images with choroidal neovascularization labeled CNV. Additionally, subfigures (a), (b), and (c) offer a visual summary of the datasets, including the number of images in each class and the division between training and test sets across all the three datasets.

required for both feature extraction and making predictions. Incorporating different techniques for feature extraction and utilizing the most commonly used image modalities in routine clinical practice ensures the comprehensiveness and generalizability of the current study. To better understand the entire process, the following subsections explore the adopted techniques for feature extraction and design and implementation of the binary classification models using PCA-LDA approach.

4.1. *Classical machine learning methods for statistical feature extraction*

The images from all three datasets, each with the size of 224 × 224 pixels, have been analyzed based on five different types of statistical image features. The mean and standard deviation of the images were computed as the first-order statistical features, based only on the pixel values and without considering the spatial relationships between the pixels.

Spatial information in the form of the texture features can be useful for the image classification. Utilizing LBP, texture analysis has been performed to facilitate the differentiation of the existing classes within each set of data. The LBP method is employed in the image processing to describe the textures by analyzing the local pixel patterns. By comparing the intensity of a central pixel in a small neighborhood with the intensity of its surrounding pixels, LBP holds a binary value based on whether each neighbor's intensity is greater or less than the central pixel. These binary values are combined to form a binary pattern, which is then converted into a decimal value and placed at the center pixel's location [44]. This process generates a new image that captures the texture information of the original image, making it useful for the tasks such as the texture analysis and the image classification. For the LBP application, we have set the radius to 1 and the number of points to 8. These parameters determine the surrounding pixels that must be included in the LBP analysis. We have also opted for the "uniform" method. This method helps to create consistent patterns and increase the robustness of the analysis against noise and fluctuations.

Additionally, we have adopted GLCM to quantify the textures via spatial pixel intensity relationships. To retrieve an image's texture statistically, the GLCM counts how often pairs of the pixels with specific intensity values occur at the same relative position throughout the image. After computing the GLCM, six statistical features, Contrast, Dissimilarity, Homogeneity, ASM (Angular Second Moment), Energy, and Correlation, can be extracted, each providing different measures of the texture, such as uniformity, contrast, and complexity. In the current analysis context, GLCM has been used with specific parameters to extract texture information from the medical images. The distances have been set from 1 to 7, representing the spatial relationships between pixel pairs, while the angles 0°, 45°, 90°, and 135° have been chosen to capture details from different directions. We have also considered multiple GLCM properties during feature extraction to ensure we capture all the essential texture aspects.

The HOG as our last choice for the statistical feature extraction, computes the gradient and orientation of edge pixels in an image. These orientations provide information about the global shape of the objects in the image. The HOG descriptor divides the image into smaller sections, computes the gradient and its orientation for each section, and then creates a histogram using these gradients and orientations for each region. This process forms the Histogram of Oriented Gradients [45]. We have selected 9 orientation bins for HOG, a standard choice that establishes a good balance between capturing enough gradient direction details and keeping the feature size manageable for processing. The cell size, where gradients have been calculated, has been set to (8, 8), which helps capture fine details without overloading the analysis. We have also used (2, 2) blocks, a typical configuration that balances performance and computational efficiency. Implementing these setups, we have successfully collected the HOG feature descriptors for the datasets under investigation.

Since we have aimed to implement DL methods with average pooling mode for feature extraction and subsequently compare ML and DL methods, we have decided to apply averaging over the extracted statistical features using LBP and HOG. Given the significantly high number of extracted features using LBP (50176 per image) and HOG (26244 per image), averaging was necessary to bring the feature counts to a more comparable level across the different extraction methods. The other methods generated a smaller number of features, making averaging unnecessary for them. Thus, averaging has been employed specifically for the LBP and HOG to maintain consistency and comparability in our analysis.

We have implemented these methods on the incorporated datasets to extract the statistical features from the medical images. In the next step the extracted features have been concatenated before developing the predictive PCA-LDA models. As shown in Table 1, each image is

TABEL 1
Number of the statistical features per image. Ten distinct methods were employed to extract the statistical features from the medical images. These features were then concatenated and used as the input for the developed classifier.

Statistical features	Mean	Standard Deviation	GLCM Contrast	GLCM Dissimilarity	GLCM Homogeneity	GLCM ASM	GLCM Energy	GLCM Correlation	LBP	HOG	Total
Features (per image)	1	1	32	32	32	32	32	32	1	36	231

converted into 231 statistical features after extraction and concatenation. Fig. 2, provides an overview of the proposed framework for the statistical feature extraction and classification.

4.2. Radiomics features

As for the next approach, we have utilized radiomics features for the image analysis and model development. Radiomics is a technique that uses advanced mathematical and computational methods to convert medical images into quantifiable data, facilitating detailed extraction and analysis of the information within the images [46]. These radiomics features capture detailed tissue and lesion characteristics, offering deeper insights into medical images. This data then can be used to address clinical problems, enhancing the ability to diagnose and assess treatment responses. The open-source Python package "pyradiomics" [39], is a tool for this purpose, offering comprehensive functionality for radiomics feature extraction from various medical image modalities. For this phase of the study, feature extraction has been performed using "pyradiomics", specifically designed for the extraction of radiomics features from the medical images. This ensures easy and reproducible extraction of radiomics features, consistent image analysis across different studies and applications [47–49].

The extracted features using radiomics method can be divided into six subcategories: first order statistical features, GLCM based features, Gray Level Run Length Matrix (GLRLM) based features [50], Gray Level Size Zone Matrix (GLSZM) based features [51], Neighboring Gray Tone Difference Matrix (NGTDM) based features [52], and Gray Level Dependence Matrix (GLDM) based features [53]. The GLCM captures statistical information about the distribution of pixel pairs in the image. The GLRLM focuses on sequences of pixels with similar intensity values, known as runs. The GLSZM quantifies gray level zones, defined as the number of connected pixels sharing the same intensity value. A wide and flat matrix indicates a homogeneous region of interest, while a narrow matrix indicates heterogeneity. The NGTDM measures the difference between a pixel's gray value and the average gray value of its surrounding neighbors within a certain distance. The GLDM quantifies the relationship between neighboring pixel values. If the difference between pixel values falls within a certain range, they are considered dependent. A high level of dependency indicates more significant relationships between neighboring pixels, leading to a more homogeneous texture in the image.

The radiomics extracted features have then been used to develop the predictive binary classification PCA-LDA models. Table 2, indicates the number of radiomics features extracted using each method embedded within the "pyradiomics" Python package.

4.3. Deep learning approach for feature extraction

Benefiting from a model which is initially trained on one task, a pre-trained network, and customizing it for a different task is the principle of the transfer learning. The intuition behind transfer learning for the image classification is that a model trained on a large and diverse dataset can effectively be served as a generic model of the visual world. Consequently, we can employ these learned feature maps without starting from scratch, avoiding the need to train a large model on an extensive dataset. In our current study, we have employed transfer learning as a primary method for the feature extraction in addition to the statistical and radiomics features. Five different pre-trained networks have been implemented to extract deep features from the medical images. VGG16, ResNet50, DenseNet121, InceptionV3, and MobileNetV2

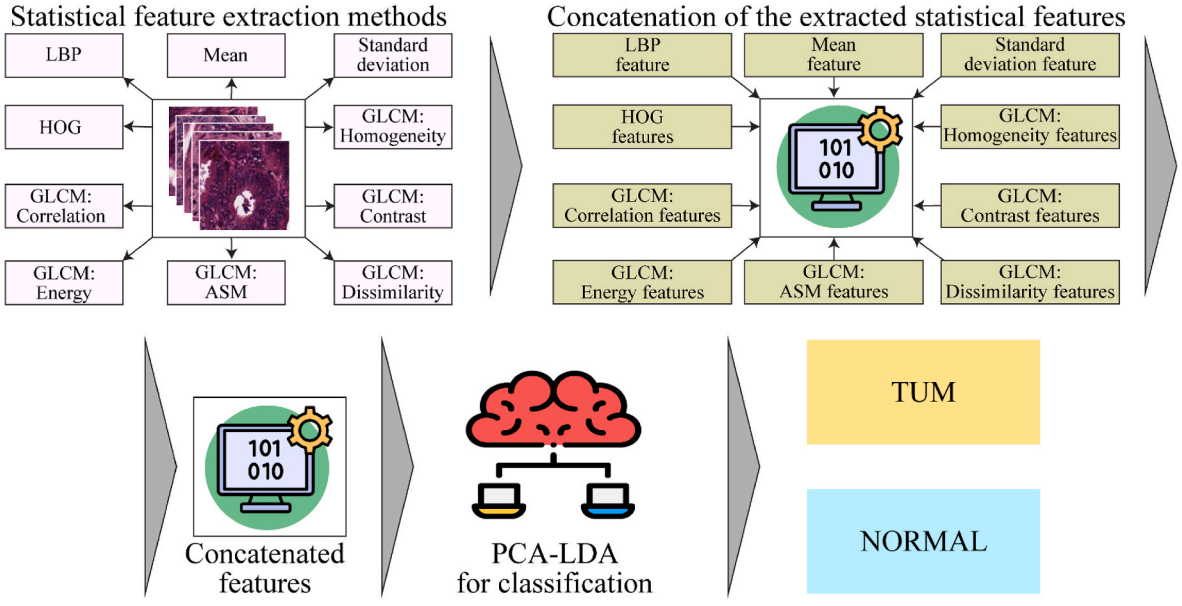


Fig. 2. *Overview of the proposed framework combining the statistical features to establish the binary classification tasks.* This figure illustrates the framework, highlighting the integration of the various methods for extracting the statistical features from the incorporated medical images within the datasets. The extracted features were always concatenated before being fed into the PCA-LDA, for the classification task. The workflow presented in this figure uses the H&E-stained dataset as an example. The same procedure was also applied to the retina OCT and chest X-ray datasets.

TABEL 2
Number of the radiomics features per image. For the radiomics features, the open-source Python package "pyradiomics" was utilized. Using six different techniques embedded within this package, 93 radiomics features were extracted from the medical images. These features were subsequently used for model development.

Radiomics features	First order statistics	GLCM	GLSZM	GLRLM	NGTDM	GLDM	Total
Features (per image)	18	24	16	16	5	14	93

with the average pooling mode have been employed as the advanced DL techniques for the feature extraction and the model development. We aimed to select the most commonly used pre-trained models in image data analysis. Each of these five networks has its unique configurations and strengths. VGG16 is known for its simplicity and depth, effectively capturing fine details. ResNet50 stands out because of its innovative use of residual connections, which help tackle the vanishing gradient problem that can occur in deep networks. DenseNet121 features dense connections between layers, allowing it to reuse features and maintain efficient gradient flow. InceptionV3 employs multiple filter sizes within the same layer to capture features at different scales. Lastly, MobileNetV2 is designed for efficiency, especially in mobile and embedded applications. It has the potential to establish a balance between performance and speed, making this model an excellent option for real-time medical image analysis. This strategy allowed us to compare the effectiveness of each pre-trained model against the others. By examining the trade-offs between accuracy, model efficiency, and the time it takes to produce results, we can identify which model, if any, stands out as the best. Table 3, displays the number of features extracted from the images using various pre-trained models as the feature extractors.

Deep learning approach	VGG16	ResNet50	DenseNe121	InceptionV3	MobileNetV2
Features (per image)	512	2048	1024	2048	1280

4.4. Model development for classification

Numerous techniques and algorithms for the data classification exist within the fields of classical ML and advanced DL. However, in this study, our primary focus lies in exploring the impact of various feature extraction methods on the classification performance rather than optimizing the predictive models. Therefore, the basic PCA-LDA model has been selected as the classifier for this research to minimize the influence of other hyperparameters during the model training. This approach enables us to focus exclusively on the effects of using different groups of features, including the statistical, radiomics, and deep features for the model development.

PCA is an unsupervised technique used to reduce the dimensionality of a dataset. It achieves this by geometrically projecting the data onto lower-dimensional spaces known as Principal Components (PCs) while still preserving critical patterns or correlations among variables without prior knowledge of the target variables. This process enables the creation of a condensed representation that retains the core essence of the original data. LDA is mainly used in supervised classification tasks to reduce dimensionality. Its core purpose is to model the distinctions between groups, effectively separating two or more classes. LDA achieves this by transforming features from a higher-dimensional space to a lower-dimensional one. While LDA shares some similarities with PCA, there are notable differences. LDA, being a supervised technique, concentrates on maximizing the distance between the means of existing classes while minimizing the variation within each class. Regarding this functionality, in the domain of ML, LDA can be designed as a supervised learning algorithm for classification. Its primary objective is to identify a linear combination of features that optimally differentiate classes within a dataset.

In this study, we combined PCA with LDA to harness the dimensionality reduction capabilities of PCA and the discrimination strength of LDA. This approach effectively transformed the data into a lower dimension and facilitated the design of the binary classification models. We thoroughly explored a range of PCA components, varying from 1 component to 100 components. Employing a train/test split cross-validation approach [54], we determined the optimal PCA component. Our training data was divided into the training and validation sets, with 80 % allocated for the training and 20 % for the validation. This entailed iterative training on the training set and validating on the validation set for each PCA component. Through this iterative process, the optimal number of PCA components that produced the highest mean sensitivity for the validation set was identified. Following this, we applied PCA using the optimal components to the training data, developed the binary predictive model, and generated predictions for the independent test set. Subsequently, mean sensitivity values have been calculated for the predictions to evaluate the performance of the classification models on the independent test set. This metric assesses the strength of the developed models in distinguishing between the existing classes within each dataset.

TABEL 3
Number of the extracted features per image using five different DL pre-trained models. By implementing these pre-trained models as the feature extractors, varying numbers of deep features were extracted from the medical images. Each pre-trained network was used as an independent extractor and analyzed separately.

Deep learning approach	VGG16	ResNet50	DenseNe121	InceptionV3	MobileNetV2
Features (per image)	512	2048	1024	2048	1280

4.5. Software

The computations were performed on a commercially available PC equipped with an AMD Ryzen Threadripper 3960X 24-Core Processor (48 threads, 3.79 GHz, 128 GB RAM). Additionally, an NVIDIA GeForce RTX 3090 GPU was utilized, featuring 10,496 CUDA cores, a base clock speed of 1440 MHz, and a boost clock speed of 1710 MHz. The GPU includes 24,576 MB of GDDR6 memory, with a memory bandwidth of 936.10 GB/s, a 384-bit memory interface, and a memory data rate of 19.50 Gbps. The analysis was conducted using the custom-written functions in Python programming language.

5. Results

ML methodology relies on processing massive amounts of data, which require significant resources, to enhance the accuracy of the predictions. Feature extraction is an efficient technique for reducing resource consumption while preserving essential information. Features are the piece of information applicable for solving specific application and to represent significant properties of images [4]. The relevance and quality of the extracted features plays a key role in determining the effectiveness of ML and DL models. Considering the significance of feature extraction, the choice of feature extractors and input features can greatly impact the accuracy of classification and the requirements for the training set. Depending on the nature of the data and the objective of

the analysis, some features might be more effective in identifying and revealing the relevant patterns and characteristics. By exploring various feature types, such as statistical, radiomics, and deep features, one can comprehensively evaluate their impact on classification performance and select the most suitable feature type for a given problem. This led us to implement a range of commonly employed feature extraction methods on the rich subset of medical images. Our goal was to study the potential effects of different feature types on the classification performance and to identify the most efficient one in terms of extraction time and classification accuracy. Ultimately, we aim to find a method that is more efficient in the real-world applications.

5.1. Radiomics and statistical features

In this section, the performance of radiomics and statistical features has been thoroughly investigated. Utilizing the radiomics and statistical features, we have developed the binary PCA-LDA models and made predictions for the independent test set. In all the cases studied, we have ensured a consistent and similar number of images within the training, validation, and test sets, enabling a fair and meaningful comparison. Fig. S1 in the supplementary materials, shows a comprehensive overview of the analysis conducted for the H&E-stained, retina OCT, and chest X-ray images. When comparing the time taken for feature extraction, both methods proved to be time-consuming. However, a comparison between the utilization of statistical and radiomics features reveals that the statistical features require more time to extract compared to the radiomics. This trend holds true across all the datasets investigated. The reason for this might be due to the extraction of a larger number of the statistical features. Additionally, "pyradiomics", an open-source Python package developed and optimized by a community of experts, ensures efficient algorithms and implementations tailored specifically for feature extraction tasks, further accelerating the extraction process. Nevertheless, given the number of features, we can still consider these two traditional approaches relatively slow. Prediction times for the independent test set in all the scenarios were quite the same, comparable and fast. When evaluating the model performance using mean sensitivity values as the employed metric, the statistical features generally outperformed the radiomics, except for the H&E-stained dataset. In the case of the H&E-stained dataset, the statistical features achieved a mean sensitivity of 90.8 %, while the radiomics features enhanced it to 94.8 %. H&E-stained images are inherently complicated due to the intricate cellular and tissue structures they present. This complexity requires a more sophisticated approach to capture subtle texture patterns, pixel intensity variations, and relations that may indicate pathological changes. Radiomics features extracted using "pyradiomics" can be considered a more well-suited methodology for H&E-stained images when compared with the statistical features. The statistical methods applied in our study, including GLCM, HOG, LBP, mean, and standard deviation, primarily capture first- and second-order statistics and edge-related information. These methods may be more effective in simpler imaging modalities, where basic texture patterns or gradient information suffice for classification. However, they might not thoroughly capture the complex spatial relationships in H&E-stained images, limiting their ability to achieve the same level of predictive accuracy as the radiomics features.

Conversely, the statistical features yielded a mean sensitivity of 94.6 % for the retina OCT images, whereas by radiomics features decreased the score to 91 %. Similarly, in the chest X-ray dataset, mean sensitivities of 93.2 % and 92.2 % were attained using the statistical and radiomics features, respectively. Despite these variations, the narrow range of changes for mean sensitivity values across all the studied cases prevents us from establishing the definitive conclusions regarding the best or worst extracted features. Considering these findings, low mean sensitivity values alongside the time estimations of the feature extraction, it seems that these extracted features might not be the optimal choices for real-world applications.

5.2. Deep learning approach for feature extraction through implementation of the pre-trained models

Pre-trained neural networks are highly capable of recognizing a wide array of patterns in images. These models can be used for feature extraction, which enables quick and effective analysis of new visual data. This section comprehensively explores the analysis results for the three incorporated datasets when utilizing various pre-trained networks as the feature extractors. The deep extracted features have been incorporated into the PCA-LDA binary classification models as the input. Fig. S2, shows that under the same conditions as for the statistical and radiomics features, deep features outperformed them in terms of feature extraction and prediction times, as well as the mean sensitivity values. Among the pre-trained models used, ResNet50 and DenseNet121 consistently demonstrated the best performance across all the studied datasets, resulting in the highest mean sensitivities. For the H&E-stained dataset (Fig. S2a), feature extraction times range from about 10 min with MobileNetV2 to 15 min with DenseNet121. Using DL models for feature extraction significantly improved the classification performance, with InceptionV3 achieving the lowest mean sensitivity of 94.2 % and ResNet50 achieving the highest at 96.9 %. Given the fast feature extraction and prediction times, along with high classification scores, these models demonstrate both efficiency and effectiveness. For the chest X-ray dataset (Fig. S2b), a similar trend to that of the H&E-stained images was observed. Mean sensitivity values range from 90.2 % to 96 %, with InceptionV3 and ResNet50 yielding the lowest and highest values, respectively, as the feature extractors. Feature extraction took approximately 15 min at most with DenseNet121 and around 10 min at the shortest with MobileNetv2. The combination of the efficient extraction times, rapid prediction times, and the enhanced model performance underscores the efficiency of the DL in this context as well. In the case of retina OCT images (Fig. S2c), the utilization of the pre-trained models demonstrated proficiency, reflecting the previous two cases. Feature extraction times ranged from nearly 15 min with DenseNet121 to about 10 min with MobileNetV2. Prediction times, and model performance were notably enhanced, with mean sensitivity ranging from 97.6 % using VGG16 as the extractor to 98.6 % with DenseNet121, proving the effectiveness of these models.

5.3. Overview of the results for the statistical, radiomics, and deep features

The final phase of the analysis involved comparing the results obtained from three different approaches: using radiomics and statistical features and employing advanced DL techniques for deep feature extraction. These three groups of features have been systematically fed into the PCA-LDA models, serving as the implemented classifier.

Fig. 3, illustrates the comparative analysis results for a single H&E-stained, chest X-ray, and retina OCT image, obtained from developing the predictive PCA-LDA models using various features that capture different aspects of the medical images. For the H&E-stained dataset, shown in Fig. 3a, the statistical and radiomics features were among the slowest to extract features of a single H&E-stained image. Implementing the pre-trained networks as the feature extractors significantly decreased the time required for feature extraction, especially with MobileNetV2. Regarding prediction time, although differences between the extracted features were minor, some variations were still evident. These variations demonstrated the superiority of the DL approach, which required less time for making predictions on a single H&E-stained image. Considering the entire extraction and prediction process, the statistical and radiomics features for model development were time-consuming and the slowest options. Model performance improved remarkably when using the features extracted with the pre-trained models and inputting them into the PCA-LDA models, except for InceptionV3.

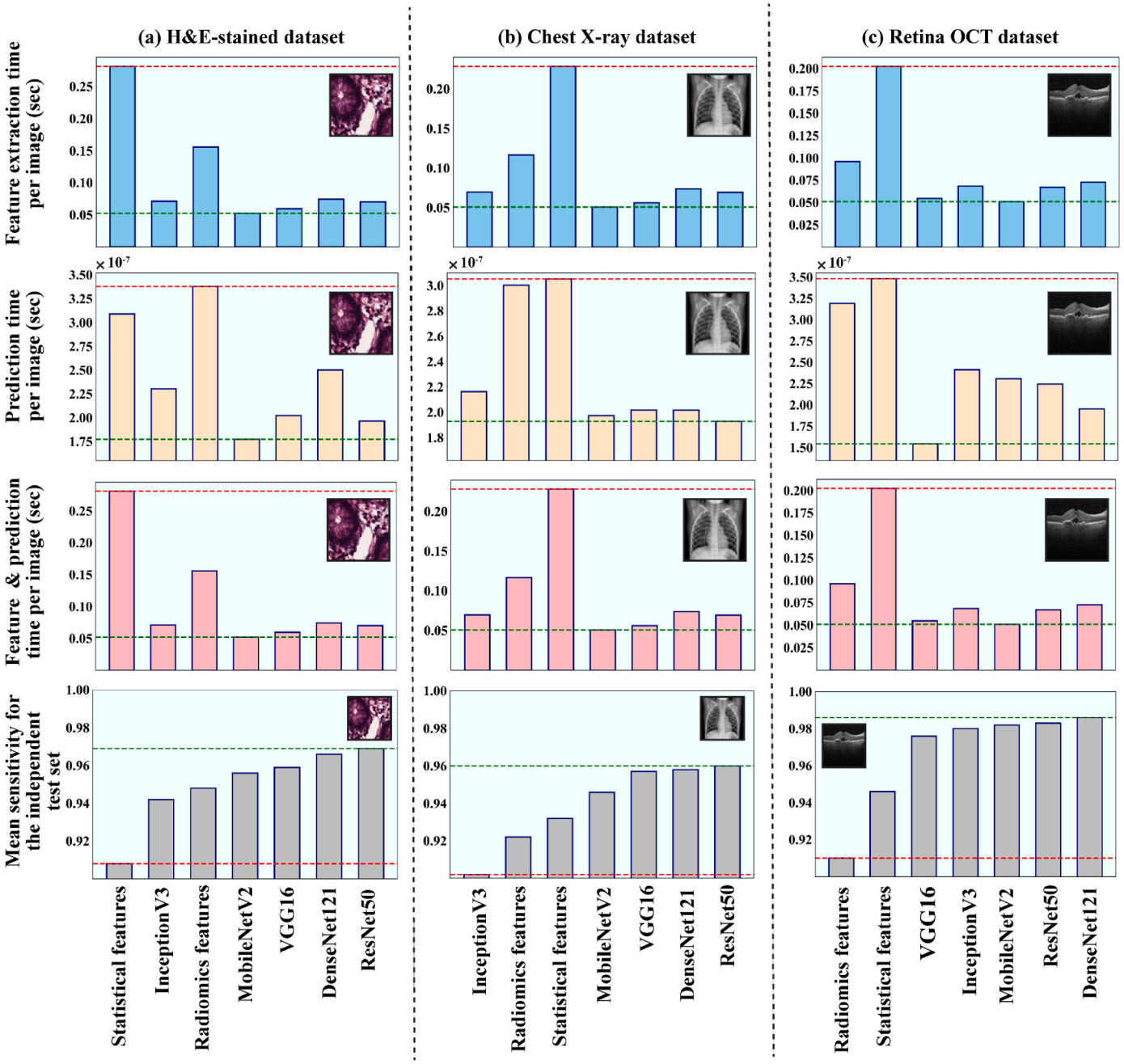
Fig. 3b, shows a similar trend for the single chest X-ray image as

observed for the single H&E-stained image. The statistical and radiomics features extraction was time-consuming compared to using the pre-trained networks for feature extraction. Although the range of prediction times was narrow, differences still existed between the methods, with the DL approach capable of predicting the single chest X-ray image in a shorter time. Considering the whole extraction and prediction process, it is clear that the statistical and radiomics features were slow both during the extraction and prediction of a single X-ray image. In contrast, the pre-trained networks significantly decreased the extraction and prediction times.

When analyzing the retina OCT dataset in Fig. 3c, it is evident that the observed patterns for the two former scenarios remained consistent. The statistical and radiomics features can still be categorized as time-consuming methods for feature extraction of a single retina OCT scan. This trend persists in the prediction times where the DL methodologies outperformed the statistical and radiomics approaches. The mean sensitivity values for the independent test set also improved notably when using the pre-trained models as the feature extractors.

In general, Fig. 3, reveals that DL significantly improved the analysis of the medical images, particularly in terms of time estimations for the feature extraction and prediction, as well as the model performance on the independent test sets. Despite some exceptions, DL not only enhances the classification results but also accelerates the process, which is crucial for disease diagnosis and treatment in real-world applications.

For a more detailed breakdown, the supplementary materials (Fig. S3) provide exact data illustrating the advantages of DL in terms of feature extraction times, prediction times, and mean sensitivity values. Furthermore, Tables S1, S2, and S3, along with Fig. S4, provide additional numerical and statistical details to complement the findings of this comparative study. These materials delve into the performance



(caption on next page)

Fig. 3. *Comparative analysis of statistical, radiomics, and deep features for extraction, prediction, and PCA-LDA model per.* (a) Results for the H&E-Stained dataset. The analysis reveals that the pre-trained models, with the exception of the InceptionV3 network, generally outperformed the statistical and radiomics features in feature extraction from the H&E-stained images. In terms of feature extraction time, the statistical features took the longest to extract for a single H&E-stained image, followed by radiomics features as the second slowest approach. The pre-trained networks demonstrated significantly greater efficiency, with MobileNetV2 being the fastest method of extraction. This improvement is particularly notable when comparing the number of the statistical, radiomics, and deep features per image (see Tables 1 and 2). For prediction time, the statistical and radiomics features were among the slowest methods, whereas the deep features, especially those extracted using MobileNetV2, proved to be faster. Considering the entire extraction process and prediction for a single image, DL approaches were far more efficient and faster compared to the statistical and radiomics features. Lastly, evaluating the performance of the developed classifier through mean sensitivity values, ResNet50, when applied as the feature extraction method, achieved the highest mean sensitivity. In contrast, the use of the statistical and radiomics features resulted in lower sensitivity scores. (b) Results for the chest X-ray dataset. In terms of feature extraction time, the statistical features required the longest time to extract from a single X-ray image, followed by the radiomics features as the second slowest approach. The pre-trained networks demonstrated significantly greater efficiency, with MobileNetV2 being the fastest extraction method. For prediction time, the statistical and radiomics features were among the slowest methods, whereas the deep features, particularly those extracted using ResNet50, proved to be faster. When considering the entire process of extraction and prediction for a single image, deep learning approaches were far more efficient and faster compared to the statistical and radiomics features. Lastly, evaluating the performance of the developed classifier through the mean sensitivity values, ResNet50, when used as the feature extraction method, achieved the highest mean sensitivity. In contrast, the use of the statistical and radiomics features resulted in lower sensitivity scores. With the exception of InceptionV3, the utilization of deep features notably improved the classification performance. (c) Results for the retina OCT dataset. In terms of feature extraction time, the statistical features required the most time to extract from a single OCT image, followed by the radiomics features as the second slowest approach. The pre-trained networks showed significantly greater efficiency, with VGG16 being the fastest extraction method. The statistical and radiomics features were among the slowest methods for prediction time, whereas the deep features, particularly those extracted using VGG16, were faster. Considering the entire process of extraction and prediction for a single image, deep learning approaches were far quicker compared to the statistical and radiomics features. When using the feature extraction method to assess the classifier's performance through the mean sensitivity values, DenseNe121 achieved the highest mean sensitivity. In contrast, using the statistical and radiomics features resulted in lower sensitivity scores. Clearly, the use of the deep features notably enhanced the classification performance.

metrics beyond mean sensitivity values and include confusion matrices for the best- and worst-performing extraction techniques across the three examined medical imaging modalities.

5.4. Computational resource analysis and model efficiency

This section provides a concise overview and comparison of the key findings from the study. We will focus on several important factors, including memory usage for the images, time estimates for feature extraction and prediction, and the performance metric of mean sensitivity values. By closely examining these aspects, we gain a clearer understanding of the study's results, assess the efficiency of each extraction method, and draw confident, well-supported conclusions.

Table 4, provides a detailed analysis of the H&E-stained dataset. This dataset consisted of 12,000 images, across different classes for the training and testing sets. When saved on disk, the total size of these images was 1.86 GB. As part of our analysis workflow, feature extraction followed data collection as the second essential step. This process significantly reduced the memory usage of the images. When InceptionV3 was used as the feature extractor, memory usage for extracted features was reduced to 93.7 MB. In comparison, employing "pyradiomics" for radiomics feature extraction resulted in the lowest memory requirement at just 8.52 MB. The statistical features required 18.2 MB, while VGG16, DenseNet121, MobileNetV2, and ResNet50 required 20.2 MB, 46.9 MB, 51.3 MB, and 88.3 MB, respectively. Considering time estimations and mean sensitivity values, the study reveals that DL models outperformed the traditional methods in handling the complexity of H&E-stained images in shorter time frames.

Upon a detailed analysis of the chest X-ray results (Table 5), a pattern consistent with that observed in the H&E-stained images emerges. Both the training and test sets of chest X-ray images (12000 images) collectively occupied 460 MB of disk space. However, following feature extraction, there was a significant reduction in storage requirements. Specifically, the extracted features compressed to just 93.7 MB when InceptionV3 and ResNet50 were used, while radiomics features consumed 8.52 MB, marking the smallest storage. The statistical features required 18.2 MB of disk space, while feature extraction with VGG16 consumed 23.4 MB. DenseNet121 followed with a space requirement of 46.9 MB, and MobileNetV2 occupied 50.6 MB. The DL methods demonstrate significant potential for disease diagnostics using x-ray images, particularly when evaluating their accuracy scores and the time efficiency of both extraction and prediction processes.

In Table 6, we present the storage requirements for the retina OCT dataset, (12000 images across two classes, divided between the training and test sets). Initially, the raw images occupied 376 MB of disk space. After extracting features using various models, the storage requirements were significantly reduced. Specifically, when using ResNet50 and InceptionV3 for feature extraction, the space usage dropped to 93.7 MB. Further breakdown shows that the statistical features occupied 18.2 MB, VGG16 extracted features required 20.2 MB, DenseNet121 utilized 46.9 MB, and MobileNetV2 needed 55.4 MB. Comparing the results from the traditional statistical and radiomics feature extraction methods further reaffirms that pre-trained networks can efficiently differentiate between existing classes in the OCT scans, taking into account both the performance and latency.

It is important to highlight that, based on our observations, feature extraction, as promised, reduces the computational costs by refining image data to the most relevant attributes for the analysis and, when appropriately selected, can also enhance the performance of classification tasks. This is particularly evident when comparing the outcomes from the statistical and radiomics features to those obtained using pre-trained DL networks for feature extraction. DL methods have demonstrated superior efficiency and accuracy in comparison to traditional approaches. However, there remain opportunities to further optimize

TABEL 4
Performance comparison of feature extraction methods on the H&E-Stained dataset. The table summarizes the time required for feature extraction and prediction, mean sensitivity, and memory usage for various feature extraction methods applied to the H&E-stained dataset.

H&E-stained dataset:Feature extraction method	Time for extraction (Per dataset (mins))	Tiem for prediction (Per dataset (msec))	Mean sensitivity (%)	Memory usage (MB)
Statistical features	56.2	0.6179	90.8	18.2
Radiomics features	31.2	0.6758	94.8	8.52
Deep features: VGG16	11.9	0.4046	95.9	20.2
Deep features: ResNet50	14	0.3932	96.9	88.3
Deep features: DenseNet121	14.9	0.5004	96.6	46.9
Deep features: MobileNetV2	10.4	0.355	95.6	51.3
Deep features: InceptionV3	14.4	0.4606	94.2	93.7

TABEL 5
Performance comparison of feature extraction methods on the chest X-ray dataset. The table summarizes the time required for feature extraction and prediction, mean sensitivity, and memory usage for various feature extraction methods applied to the chest x-ray dataset.

Chest X-ray dataset:Feature extraction method	Time for extraction (Per dataset (mins))	Tiem for prediction (Per dataset (msec))	Mean sensitivity (%)	Memory usage (MB)
Statistical features	45.68	0.6105	93.2	18.2
Radiomics features	23.35	0.6008	92.2	8.52
Deep features: VGG16	11.24	0.4035	95.7	23.4
Deep features: ResNet50	13.82	0.3857	96	93.7
Deep features: DenseNet121	14.74	0.4034	95.8	46.9
Deep features: MobileNetV2	10.14	0.3948	94.6	50.6
Deep features: InceptionV3	13.9	0.4326	90.2	93.7

time estimation for radiomics and statistical features by adopting strategies such as parallel computing. While we currently do not have access to such resources, our study's findings suggest that even with faster processing times for these features, the performance of classification tasks remains unsatisfactory. In other words, accelerating the process does not improve the quality or strength of the features, nor does it lead to better results. Given these considerations for the medical image analysis, pre-trained networks within DL frameworks emerge as the more efficient and accurate option for feature extraction. These networks offer significant potential for integration into routine practices, providing robust performance for future applications.

6. Discussion

This study was designed to systematically explore various feature extraction techniques for the medical images to identify the best-performing approach based on two significant factors: the time required to complete the task, including feature extraction and prediction for an independent test set, and the main performance metric, mean sensitivity. To achieve the objective of the analysis, we selected three of the most widely used imaging modalities in healthcare: H&E-stained images, retina OCT, and chest X-ray images. These served as the inputs for the developed binary predictive models. We applied no pre-processing to the image data and implemented a basic PCA-LDA classifier to minimize external influences on the analysis. This approach allowed us to isolate and examine the impact of various feature groups in the classification tasks. We tested three primary categories of features: statistical, radiomics, and DL based.

Tabel 6
Performance comparison of feature extraction methods on the retina OCT dataset. The table summarizes the time required for feature extraction and prediction, mean sensitivity, and memory usage for various feature extraction methods applied to the retina OCT dataset.

Retina OCT dataset:Feature extraction method	Time for extraction (Per dataset (mins))	Tiem for prediction (Per dataset (msec))	Mean sensitivity (%)	Memory usage (MB)
Statistical features	40.48	0.6965	94.6	18.2
Radiomics features	19.4	0.6387	91	8.52
Deep features: VGG16	10.9	0.3088	97.6	23.4
Deep features: ResNet50	13.42	0.4491	98.3	93.7
Deep features: DenseNet121	14.53	0.3908	98.6	46.9
Deep features: MobileNetV2	10.2	0.4616	98.2	50.6
Deep features: InceptionV3	13.68	0.4829	98	93.7

The results indicate that employing pre-trained DL networks for feature extraction reduced the analysis time substantially in comparison to the traditional statistical and radiomics approaches. For all the modalities studied, feature extraction required less than 15 min for all DL models, with an average of around 10 min for MobileNetV2 and VGG16, 13 min for InceptionV3 and ResNet50, and 14 min for DenseNet121. In contrast, the traditional methods were much slower, with statistical feature extraction averaging 50 min and radiomics taking about 24 min. This substantial reduction in the processing time with DL approaches allows for quicker diagnostic decision-making, which is essential for time-critical clinical situations such as emergency care.

In addition, when transitioning from the traditional methods to DL, there was a significant enhancement in the classification performance. Using the statistical and radiomics features as the inputs to the classifier, the mean sensitivity across all the studied cases averaged around 92 %. However, utilizing DL significantly boosted the classification performance: InceptionV3 achieved an average mean sensitivity of 94 %, MobileNetV2 increased the score to 96 %, and VGG16 reached approximately 96.4 %. DenseNet121 and ResNet50 further elevated the average classification score to 97 %. These improvements demonstrate that DL-based models can more effectively distinguish between disease conditions, thereby enhancing the accuracy of early disease diagnosis process. Higher sensitivity reduces the likelihood of false negatives, which is crucial for conditions such as cancer or pneumonia, where early and precise diagnosis can dramatically influence patient outcomes.

These discoveries reinforce the value of DL models not only for their superior accuracy but also for their efficiency in handling complex datasets, positioning them as a preferred approach in medical diagnostics where rapid and reliable results are crucial.

In the following subsections, we will discuss the results for each dataset individually to provide a clearer, more detailed analysis.

6.1. Analysis results: Hematoxylin and Eosin stain (H&E-stained) dataset

In this study, H&E-stained images, the most complex image data type analyzed, achieved the highest mean sensitivity scores when classified with DL approach. Using ResNet50 as the feature extractor allowed precise differentiation between the normal and tumorous tissues, reaching a maximum accuracy of 96.9 %. In contrast, models trained on the statistical features showed the lowest performance, with sensitivity dropping to 90.8 %. Processing the statistical features was also the most time-consuming, taking even longer than the radiomics feature extraction, which was the second slowest. Pre-trained DL networks proved highly efficient, completing tasks in under 15 min, whereas the statistical and radiomics methods required approximately 1 h and 30 min each.

Overall, the performance metric for the H&E-stained images increased from 90.8 % to 96.9 % implementing the DL technique, demonstrating clear superiority over the radiomics and statistical methods in accuracy, efficiency, and reliability. These results highlight the effectiveness of the DL approach and its strong potential for application in complex image-based diagnostics.

6.2. Analysis results: Chest X-ray dataset

The next dataset studied was chest X-ray images which are commonly used in the field of radiology to detect respiratory conditions, such as pneumonia. The images were categorized into two classes: healthy (normal) and those showing signs of pneumonia. Surprisingly, when InceptionV3, a DL based feature extraction model, was applied, it

yielded a relatively low mean sensitivity score of 90.2 %. Several factors could explain this outcome. The complex architecture of InceptionV3, designed with a multi-branch structure to capture features at multiple scales, might inadvertently complicate feature extraction in the chest X-ray images. This complexity could result in suboptimal feature representation, which in turn could lead to the reduced accuracy. Additionally, the filters learned by InceptionV3 as a pre-trained model may not align well with the specific patterns present in chest X-rays, further contributing to its poor performance in detecting pneumonia. Following InceptionV3, the use of statistical and radiomics features yielded the second- and third-lowest mean sensitivity values, at 92.2 % and 93.2 %, respectively. The highest mean sensitivity, 96 %, was achieved using ResNet50 for feature extraction and classification. Densenet121, VGG16, and MobileNetV2 ranked second, third, and fourth in terms of mean sensitivity scores.

Similar to our observations with the H&E-stained dataset, the statistical and radiomics feature extraction required the longest processing times, while DL methods demonstrated both speed and accuracy, highlighting their potential for real-world applications.

6.3. Analysis results: Optical Coherence Tomography (OCT) dataset

In the third dataset, retinal OCT images were categorized into two classes: healthy retinas without fluid or edema (NORMAL) and retinas with choroidal neovascularization (CNV). The traditional radiomics and statistical feature extraction methods yielded the lowest results, achieving accuracy rates of only 91 % and 94.6 %, respectively, and were recognized by lengthy processing times. In contrast, the deployed pre-trained DL networks delivered notably higher accuracy values. DenseNet121 led the group with an impressive accuracy of 98.6 %, followed closely by ResNet50 at 98.3 %, with MobileNetV2 and InceptionV3 at 98.2 % and 98 %, respectively, and VGG16 at 97.6 %. Beyond achieving the excellent sensitivity scores, DL models exhibited remarkable efficiency, completing tasks including extraction and making predictions, far more rapidly than both the radiomics and statistical methods. The significant advantages in speed and reliability underscore DL models as exceptional tools for medical image analysis, further validating our observations from the two previous imaging modalities.

However, several limitations remain when implementing DL-based methods in healthcare. One major challenge is the need for extensive computational resources, such as high-performance GPUs, which are often inaccessible in resource-limited healthcare settings. Furthermore, the effectiveness of DL techniques relies heavily on the availability of large, high-quality datasets for training, which may not be feasible for rare diseases or uncommon conditions. Another concern is that DL models are often perceived as "black boxes" making it difficult to interpret the reasoning behind their decisions, a critical issue in clinical environments where transparency is essential for building clinician trust. Overcoming these challenges is necessary to fully harness DL's potential, making it possible to integrate these methods into primary healthcare settings in ways that are practical, interpretable, and broadly accessible.

7. Key findings

The principal findings of this study can be briefly outlined as:

- Our study aimed to assess the impact of different features, such as the statistical, radiomics, and deep features, on the performance of the binary classification model.
- Our research findings have shown that the radiomics and statistical features utilized in our study are time-intensive when it comes to feature extraction from the three integrated datasets.
- Based on the mean sensitivity values, our evaluation metric, the radiomics and statistical features have demonstrated inadequate applicability for real-world scenarios due to their high latency, and low performance across all the three integrated datasets.
- Through advanced DL approach, utilizing the pre-trained networks as the feature extractors has been proven to be fast and efficient.
- Across all the implemented pre-trained networks used as the feature extractors, DenseNet121 consistently took approximately 15 min to extract features, whereas MobileNetV2 was the fastest method, taking around 10 min.
- Considering the analysis results for the three incorporated datasets, utilization of the deep features for model development consistently outperformed the statistical and radiomics features.
- Among the pre-trained models used as the feature extractors across the three datasets, ResNet50 and DenseNet121 ranked among the best performing models, achieving the highest mean sensitivity scores.
- Regarding the analysis criteria, including time estimations for feature extraction and prediction, as well as the model performance in binary classification tasks, DL proved to be a superior approach compared to the implementation of the statistical and radiomics features.

8. Conclusion

In this comparative investigation, we examined and analyzed the impact of different feature extraction methods on the performance of binary classification models for the most applicable medical images in clinical practices. To enhance the generalizability of the study, we selected three widely used image modalities from routine healthcare practice: H&E-stained datasets (Pathology), chest X-ray images (Radiology), and retina OCT (Ophthalmology) images. Three distinct feature extraction methods were applied to these modalities. The first set of statistical features was obtained using standard techniques, including mean, standard deviation, LBP, GLCM, and HOG. For the second approach, an open-source Python package, "pyradiomics" was employed to extract radiomics features. The third methodology involved utilizing advanced DL techniques. Five pre-trained networks, VGG16, ResNet50, DenseNet121, MobileNetV2, and InceptionV3, were used as the feature extractors to capture deep and complex patterns from the medical images.

The extracted features were used as input for the PCA-LDA models, which opted as the binary classifiers for the classification tasks. Using a train/test split cross-validation method, the performance of the models was assessed based on two evaluation metrics: latency and performance. Latency measured the time taken by the model to process and complete the extraction and prediction. Performance evaluation has been evaluated based on the mean sensitivity values of the classification models.

Our findings highlight the significance of feature extraction methods in the context of the medical image analysis. It was demonstrated that the utilization of the statistical and radiomics features might not offer the most effective solutions for improving the classification outcomes. Considering the time required for the feature extraction, these methods proved to be slow and inefficient, resulting in unsatisfactory mean sensitivity values, particularly when considering their potential applications in the real-world scenarios.

Within the context of transfer learning and DL application, the H&E-stained dataset demonstrated its lowest mean sensitivity value at 94 % when InceptionV3 was utilized as the feature extractor. However, this value has significantly improved to 96.9 % with the utilization of ResNet50. The chest X-ray images showed the mean sensitivity increase from 90.2 % with InceptionV3 to 96 % with ResNet50. In the case of retina OCT images, features from the VGG16 implementation resulted in the lowest mean sensitivity value of 97.6 %, while DenseNet121 delivered the highest score of 98.6 %.

A consistent pattern was also observed across all three datasets when comparing latency for various feature extraction methods. The feature extraction times decreased significantly when transitioning from the

statistical and radiomics features to the pre-trained models. Considering only DL methods and all the incorporated datasets, DenseNet121 consistently took around 15 min to extract features, while MobileNetV2, the fastest model, took nearly 10 min, which is significantly faster than the traditional extraction methods. Furthermore, the importance of these features becomes even more pronounced when looking into their quantities: 231 features were extracted using the statistical methods, 93 were obtained through the radiomics, and DL methods generated an impressive range of 512 to 2048 features. Remarkably, the DL methods not only produced significantly more features but also did so in considerably shorter extraction times compared to the lengthier processes required by the other two approaches.

Prediction time was another evaluation criterion. Although prediction times were fast and efficient across all the datasets and incorporated feature types, slight differences were noted. The models trained on the statistical and radiomics features required more time to predict the independent test set, whereas in the case of the use of the pre-trained models as the feature extractors models has proven to be slightly faster in prediction phase. Except for the InceptionV3, the utilization of pre-trained networks to extract complex patterns from the medical images consistently proved to be effective across all the datasets. Employing VGG16, MobileNetv2, DenseNet121, and ResNet50 led to the improved mean sensitivity values, indicating better class differentiation within each dataset.

All in all, implementing classical ML approaches for extracting statistical features were the least efficient methodology. The analysis process was significantly slower compared to the radiomics and DL feature extraction methods across nearly all the datasets investigated. These classical techniques often require analyzing spatial relationships between pixel values and aggregating the information across entire images, making the process computationally demanding. On the other hand, radiomics and, more importantly, DL strategies efficiently decreased the computational burden through the use of optimized algorithms and pre-trained models for extracting radiomics and high-level features, respectively. The inefficiency of the statistical features affected more than just the time estimates for feature extraction and predictions. It was also reflected in the classification results, with low mean sensitivity values. These remarks highlight that statistical methods may be less efficient, especially with large-scale datasets.

Based on the detailed and comprehensive observations throughout this investigation, we concluded that the employed feature extraction strategies revealed clear trends in latency and performance for the statistical, radiomics, and deep features. Across all the imaging modalities, the statistical and radiomics features consistently required the longest extraction times, while DL methods demonstrated markedly faster, more reliable, robust, and accurate performance. Although combining the radiomics and statistical features could potentially enhance classification accuracy, the extensive time required for their extraction limits their practical use, particularly given their modest contribution to the overall model performance. Therefore, considering the time efficiency and accuracy associated with each extraction method, it is unlikely that combining these traditional methods would add significant value to the results. In contrast, DL methods emerge as the most efficient and practical solution for real-world applications requiring high accuracy and rapid processing.

9. Limitations

We have obtained reliability by comparing the efficacy of various feature extraction methods such as statistical, radiomics, and DL approaches for classifying medical images. Even though the current research is comprehensive, there are still some limitations that need to be considered. The datasets we used H&E-stained colorectal cancer tissue images, which pathologists rely on for routine diagnostics; chest X-ray images, essential for identifying fractures, abnormalities, and respiratory conditions like pneumonia; and retina OCT scans, a critical tool in diagnosing ocular diseases, represent just a small yet rich subset of the many medical image types seen in clinical practice. In our analysis, we utilized multiple datasets and methodologies to derive conclusions based on robust observations. It is important to recognize that these findings may not be universally applicable to all imaging techniques. Each imaging modality contains unique attributes, such as differing levels of noise, structural complexity, and overall texture affecting the final output.

The success of particular feature extraction methods can be significantly impacted by a variety of factors. This implies that techniques that prove effective for one form of imaging, like X-ray or H&E-stained scans, may be less successful for another type due to differing imaging conditions, noise level, or overall complexity. However, we believe that while the specific resulting scores may vary depending on the task or imaging type, the general trends of our study are likely to be applicable in similar contexts. It's not feasible to test all possible imaging modalities. The results will need verification using the methods detailed in this research for different tasks or imaging types.

We intentionally opted for the simple and basic PCA-LDA model as the fundamental procedure for conducting the classification tasks. The only varying factor during the model development was the number of PCA components, which we optimized using cross-validation. Our goal was to minimize the influence of external factors and focus mainly on the effects of different feature types, rather than on tuning the hyperparameters of the employed classifier. However, our choice of the classifier was carefully considered and justified based on the specific study objectives, using only a single model for classification may limit the ability to fully interpret the performance of each feature extraction method, as it reflects the results of the adopted classifier. Exploring alternative algorithms for establishing classification tasks could reveal other aspects of feature extraction techniques. Future studies could incorporate a range of classification models for a more in-depth interpretation of how various features interact with different models.

Researchers might also face limitations in implementing the DL approach due to hardware constraints and configuration challenges, particularly the need for access to high-performance GPUs. Although DL models deliver exceptional performance, these requirements could limit reproducibility and practical application in low-resource settings. In cases of rare diseases or uncommon medical conditions, obtaining enough high-quality data to develop DL models effectively poses a significant challenge. The need for more data can limit the model's ability to generalize and perform accurately, restricting its effectiveness. Furthermore, DL models are frequently seen as "black boxes" meaning their decision-making processes are usually unclear and difficult to interpret. This lack of transparency is particularly concerning in clinical settings, where understanding the reasoning behind a model's predictions is essential for encouraging trust among healthcare providers.

Recognizing these limitations and the logic behind them not only helps to better understand the essence of the study but also creates opportunities for potential future studies. It allows for exploring new strategies, from experimenting with different data types and trying new model development approaches to refining evaluation methods. Expanding the outlook on these areas would provide a more comprehensive interpretation of the optimal strategies for different medical imaging tasks, eventually enhancing real-world diagnostic applications.

CRediT authorship contribution statement

Pegah Dehbozorgi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Oleg Ryabchykov:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Thomas W. Bocklitz:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Ethical Statement

The following criteria are met:

1) This material is the author's own original work, which has not been previously published elsewhere.
2) The paper is not currently being considered for publication elsewhere.
3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
5) The article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.
6) All authors have been personally and actively involved in substantial work leading to the paper and will take public responsibility for its content.
7) If accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright-holder.

We agree with the above statements and declare that this submission follows the policies of Computers in Biology and Medicine as outlined in the Guide for Authors and in the Ethical Statement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the BMBF, funding program Photonics Research Germany (13N15710 (LPI-BT3)) and is integrated into the Leibniz Center for Photonics in Infection Research (LPI). The LPI initiated by Leibniz-IPHT, Leibniz-HKI, Friedrich Schiller University Jena and Jena University Hospital is part of the BMBF national roadmap for research infrastructures. The project is co-funded by the European Union (ERC, STAIN-IT, 101088997). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2025.109768.

References

[1] K. Ng, O. Faust, V. Sudarshan, S. Chattopadhyay, Data overloading in medical imaging: emerging issues, challenges and opportunities in efficient data management, J. Med. Imaging Health Inform. 5 (2015) 755–764, https://doi.org/10.1166/jmihi.2015.1449.
[2] P. Kora, C.P. Ooi, O. Faust, U. Raghavendra, A. Gudigar, W.Y. Chan, K. Meenakshi, K. Swaraja, P. Plawiak, U.R. Acharya, Transfer learning techniques for medical image analysis: a review, Biocybern. Biomed. Eng. 42 (2022) 79–107, https://doi.org/10.1016/j.bbe.2021.11.004.
[3] H. Zhang, Y. Qie, Applying deep learning to medical imaging: a review, Appl. Sci. 13 (2023) 10521, https://doi.org/10.3390/app131810521.
[4] I. Guyon, A. Elisseeff, Intro. Feature Extract. Feature Extract.: Found. Appl. (2006) 1–25, https://doi.org/10.1007/978-3-540-35488-8_1. Springer.
[5] P. Dehbozorgi, O. Ryabchykov, T. Bocklitz, A systematic investigation of image pre-processing on image classification, IEEE Access (2024), https://doi.org/10.1109/ACCESS.2024.3395063.
[6] J.N. Kather, N. Halama, A. Marx, 100,000 histological images of human colorectal cancer and healthy tissue, Zenodo10 5281 (2018) 6, https://doi.org/10.5281/zenodo.1214455.
[7] D. Kermany, K. Zhang, M. Goldbaum, Labeled optical coherence tomography (oct) and chest x-ray images for classification, Mendeley Data 2 (2018) 651, https://doi.org/10.17632/rscbjbr9sj.2.
[8] H. Abdi, L.J. Williams, Principal component analysis, Wiley Interdis. Rev.: Comput. Stat. 2 (2010) 433–459, https://doi.org/10.1002/wics.101.
[9] A. Tharwat, T. Gaber, A. Ibrahim, A.E. Hassanien, Linear discriminant analysis: a detailed tutorial, AI Commun. 30 (2017) 169–190, https://doi.org/10.3233/AIC-170729.
[10] D. Kumar, Feature extraction and selection of kidney ultrasound images using GLCM and PCA, Procedia Comput. Sci. 167 (2020) 1722–1731, https://doi.org/10.1016/j.procs.2020.03.382.
[11] R.M. Haralick, K. Shanmugam, I.H. Dinstein, Textural features for image classification, IEEE Trans. Sys. Man Cyber. (1973) 610–621, https://doi.org/10.1109/TSMC.1973.4309314.
[12] S. Agatonovic-Kustrin, R. Beresford, Basic concepts of artificial neural network (ANN) modeling and its application in pharmaceutical research, J. Pharmaceut. Biomed. Anal. 22 (2000) 717–727, https://doi.org/10.1016/S0731-7085(99)00272-1.
[13] G. Dheepak, D. Vaishali, Brain tumor classification: a novel approach integrating GLCM, LBP and composite features, Front. Oncol. 13 (2024) 1248452, https://doi.org/10.3389/fonc.2023.1248452.
[14] D.-C. He, L. Wang, Texture unit, texture spectrum, and texture analysis, IEEE Trans. Geosci. Rem. Sens. 28 (1990) 509–512, https://doi.org/10.1109/TGRS.1990.572934.
[15] L. Wang, D.-C. He, Texture classification using texture spectrum, Pattern Recogn. 23 (1990) 905–910, https://doi.org/10.1016/0031-3203(90)90135-8.
[16] C. Cortes, V. Vapnik, Support-vector networks, Mach. Learn. 20 (1995) 273–297, https://doi.org/10.1007/BF00994018.
[17] L.K. Kumari, B.N. Jagadesh, A robust feature extraction technique for breast cancer detection using digital mammograms based on advanced GLCM approach, EAI Endorsed Trans. Pervasive Health Tech. 8 (2022), https://doi.org/10.4108/eai.11-1-2022.172813 e3-e3.
[18] A.M. Reza, Realization of the contrast limited adaptive histogram equalization (CLAHE) for real-time image enhancement, J. VLSI Signal Process Sys. Signal Image Video Tech. 38 (2004) 35–44, https://doi.org/10.1023/B:VLSI.0000028532.53893.82.
[19] T. Chen, T. He, M. Benesty, V. Khotilovich, Y. Tang, H. Cho, K. Chen, R. Mitchell, I. Cano, T. Zhou, Xgboost: extreme gradient boosting, R Package Version 0 1 (4–2) (2015) 1–4.
[20] K. Kaplan, Y. Kaya, M. Kuncan, H.M. Ertunç, Brain tumor classification using modified local binary patterns (LBP) feature extraction methods, Med. Hypotheses 139 (2020) 109696, https://doi.org/10.1016/j.mehy.2020.109696.
[21] L.E. Peterson, K-nearest neighbor, Scholarpedia 4 (2009) 1883.
[22] T.K. Ho, The random subspace method for constructing decision forests, IEEE Trans. Pattern Anal. Mach. Intell. 20 (1998) 832–844, https://doi.org/10.1109/34.709601.
[23] G.I. Webb, J.R. Boughton, Z. Wang, Not so naive Bayes: aggregating one-dependence estimators, Mach. Learn. 58 (2005) 5–24, https://doi.org/10.1007/s10994-005-4258-6.
[24] A. Stefano, Challenges and limitations in applying radiomics to PET imaging: possible opportunities and avenues for research, Comput. Biol. Med. 179 (2024) 108827, https://doi.org/10.1016/j.compbiomed.2024.108827.
[25] Q.-q. Xu, W.-l. Shan, Y. Zhu, C.-c. Huang, S.-y. Bao, L.-l. Guo, Prediction efficacy of feature classification of solitary pulmonary nodules based on CT radiomics, Eur. J. Radiol. 139 (2021) 109667, https://doi.org/10.1016/j.ejrad.2021.109667.
[26] M.A. Rasool, S. Ahmed, U. Sabina, T.K. Whangbo, KONet: towards a weighted ensemble learning model for knee osteoporosis classification, IEEE Access (2024), https://doi.org/10.1109/ACCESS.2023.3348817.
[27] W. Han, L. Qin, C. Bay, X. Chen, K.-H. Yu, N. Miskin, A. Li, X. Xu, G. Young, Deep transfer learning and radiomics feature prediction of survival of patients with high-grade gliomas, Am. J. Neuroradiol. 41 (2020) 40–48, https://doi.org/10.3174/ajnr.A6365.
[28] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition, arXiv preprint arXiv:1409.1556 (2014), https://doi.org/10.48550/arXiv.1409.1556.
[29] N. Ali, J. Kirchhoff, P.I. Onoja, A. Tannert, U. Neugebauer, J. Popp, T. Bocklitz, Predictive modeling of antibiotic susceptibility in E. Coli strains using the U-Net network and one-class classification, IEEE Access 8 (2020) 167711–167720, https://doi.org/10.1109/ACCESS.2020.3022829.
[30] S.U.R. Khan, M. Zhao, S. Asif, X. Chen, Hybrid-NET: a fusion of DenseNet169 and advanced machine learning classifiers for enhanced brain tumor diagnosis, Int. J. Imag. Syst. Technol. 34 (2024) e22975, https://doi.org/10.1002/ima.22975.
[31] M. Tan, Q. Le, Efficientnet: rethinking model scaling for convolutional neural networks. International Conference on Machine Learning, PMLR, 2019, pp. 6105–6114, https://doi.org/10.48550/arXiv.1905.11946.
[32] S.U.R. Khan, S. Asif, Oral cancer detection using feature-level fusion and novel self-attention mechanisms, Biomed. Signal Process Control 95 (2024) 106437, https://doi.org/10.1016/j.bspc.2024.106437.
[33] S.U.R. Khan, S. Asif, O. Bilal, S. Ali, Deep hybrid model for Mpox disease diagnosis from skin lesion images, Int. J. Imag. Syst. Technol. 34 (2024) e23044, https://doi.org/10.1002/ima.23044.
[34] K. de Haan, Y. Zhang, J.E. Zuckerman, T. Liu, A.E. Sisk, M.F. Diaz, K.-Y. Jen, A. Nobori, S. Liou, S. Zhang, Deep learning-based transformation of H&E stained

tissues into special stains, Nat. Commun. 12 (2021) 1–13, https://doi.org/10.1038/s41467-021-25221-2.
[35] W.C. Röntgen, On a new kind of rays, Science 3 (1896) 227–231, https://doi.org/10.1126/science.3.59.227.
[36] M. Berger, Q. Yang, A. Maier, Medical Imaging Systems: an Introductory Guide, vol. 1, SpringerOpen, Gewerbestrasse, 2018, p. 263, https://doi.org/10.1007/978-3-319-96520-8.
[37] D. Huang, E.A. Swanson, C.P. Lin, J.S. Schuman, W.G. Stinson, W. Chang, M. R. Hee, T. Flotte, K. Gregory, C.A. Puliafito, Optical coherence tomography, Science 254 (1991) 1178–1181, https://doi.org/10.1126/science.1957169.
[38] N. Dalal, B. Triggs, Histograms of oriented gradients for human detection. 2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05), Ieee, 2005, pp. 886–893, https://doi.org/10.1109/CVPR.2005.177.
[39] J.J. Van Griethuysen, A. Fedorov, C. Parmar, A. Hosny, N. Aucoin, V. Narayan, R. G. Beets-Tan, J.-C. Fillion-Robin, S. Pieper, H.J. Aerts, Computational radiomics system to decode the radiographic phenotype, Cancer Res. 77 (2017) e104–e107, https://doi.org/10.1158/0008-5472.CAN-17-0339.
[40] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778, https://doi.org/10.1109/CVPR.2016.90.
[41] G. Huang, Z. Liu, L. Van Der Maaten, K.Q. Weinberger, Densely connected convolutional networks. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 4700–4708, https://doi.org/10.48550/arXiv.1608.06993.
[42] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, Mobilenetv2: inverted residuals and linear bottlenecks. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 4510–4520, https://doi.org/10.1109/CVPR.2018.00474.
[43] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826, https://doi.org/10.1109/CVPR.2016.308.
[44] T. Ojala, M. Pietikäinen, D. Harwood, A comparative study of texture measures with classification based on featured distributions, Pattern Recogn. 29 (1996) 51–59, https://doi.org/10.1016/0031-3203(95)00067-4.
[45] U. Sajid, R.A. Khan, S.M. Shah, S. Arif, Breast cancer classification using deep learned features boosted with handcrafted features, Biomed. Signal Process Control 86 (2023) 105353, https://doi.org/10.1016/j.bspc.2023.105353.
[46] M.E. Mayerhoefer, A. Materka, G. Langs, I. Häggström, P. Szczypiński, P. Gibbs, G. Cook, Introduction to radiomics, J. Nucl. Med. 61 (2020) 488–495, https://doi.org/10.2967/jnumed.118.222893.
[47] K. Bera, N. Braman, A. Gupta, V. Velcheti, A. Madabhushi, Predicting cancer outcomes with radiomics and artificial intelligence in radiology, Nat. Rev. Clin. Oncol. 19 (2022) 132–146, https://doi.org/10.1038/s41571-021-00560-7.
[48] J.E. Van Timmeren, D. Cester, S. Tanadini-Lang, H. Alkadhi, B. Baessler, Radiomics in medical imaging—"how-to" guide and critical reflection, Insights Into Imaging 11 (2020) 91, https://doi.org/10.1186/s13244-020-00887-2.
[49] R.S. Vanguri, J. Luo, A.T. Aukerman, J.V. Egger, C.J. Fong, N. Horvat, A. Pagano, J. d.A.B. Araujo-Filho, L. Geneslaw, H. Rizvi, Multimodal integration of radiology, pathology and genomics for prediction of response to PD-(L) 1 blockade in patients with non-small cell lung cancer, Nature Cancer 3 (2022) 1151–1164, https://doi.org/10.1038/s43018-022-00416-8.
[50] M.M. Galloway, Texture analysis using gray level run lengths, Comput. Graph. Image Process. 4 (1975) 172–179, https://doi.org/10.1016/S0146-664X(75)80008-6.
[51] G. Thibault, J. Angulo, F. Meyer, Advanced statistical matrices for texture characterization: application to cell classification, IEEE (Inst. Electr. Electron. Eng.) Trans. Biomed. Eng. 61 (2013) 630–637, https://doi.org/10.1109/TBME.2013.2284600.
[52] M. Amadasun, R. King, Textural features corresponding to textural properties, IEEE Trans. Sys. Man Cyber. 19 (1989) 1264–1274, https://doi.org/10.1109/21.44046.
[53] C. Sun, W.G. Wee, Neighboring gray level dependence matrix for texture classification, Comput. Vis. Graph Image Process 23 (1983) 341–352, https://doi.org/10.1016/0734-189X(83)90032-4.
[54] M.W. Browne, Cross-validation methods, J. Math. Psychol. 44 (2000) 108–132, https://doi.org/10.1006/jmps.1999.1279.

Pegah Dehbozorgi received the B.Sc. and M.Sc. degrees in Nuclear Physics and Photonics from the University of Mazandaran and the Institute for Advanced Studies in Basic Sciences, Zanjan, in 2011 and 2015, respectively. She is currently pursuing the Ph.D. degree with the Research Group of Photonic Data Science, University of Jena, supervised by Prof. Dr. Thomas Bocklitz. Her research interests lie at the intersection of image analysis and machine learning methods, driving innovation in fields like computer vision and medical diagnostics.

Oleg Ryabchykov received his B.Sc. in Applied Physics from V.I. Vernadsky Taurida National University and his M.Sc. in High Technology from Taras Shevchenko National University of Kyiv. In 2019, he successfully completed his Ph.D. at Friedrich Schiller University Jena. Currently, he works as a postdoctoral researcher, as well as a data scientist and software developer. His research is dedicated to advancing spectral data processing automation, exploring the realms of data fusion, and employing machine learning methods for photonics data to automate diagnostics in the medical and biological fields.

Thomas Bocklitz received the Diploma degree in theoretical physics, in 2007, the Ph.D. degree in chemometrics, in 2011, and the degree in physics and mathematics from Friedrich Schiller University Jena. He is head of a research department "Photonic Data Science" at the Leibniz-Institut of Photonic Technology and runs a research group at the Institute of Physical Chemistry. He is fascinated by the fact that problems can be understood and solved using mathematical methods. His research interests include machine learning for photonic image data, chemometrics/machine learning for spectral data, correlation of different measurement methods and data fusion.